ALBERT YOSHIMOTO

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EDUCATION

University of Texas at El Paso (UTEP) El Paso, TX
M.S. in Artificial Intelligence — GPA: 4.0 *January 2024 – Expected May 2027*

Relevant Coursework: Decision Making in AI, Advanced Algorithms, Theory of Computation, Computer Security, Computer Networks, Secure Web-Based Systems (PHP, JavaScript, MySQL), Software Architecture

New Mexico State University Las Cruces, NM
B.M. in Instrumental Performance — GPA: 3.85 *August 2019 – May 2023*

RESEARCH EXPERIENCE

Army HBCU-MI SPARK Summer 2026
Research Fellow

- Selected for the Army HBCU-MI SPARK fellowship to conduct CyberAI research at UTEP.
- Co-developing a Capture-the-Flag (CTF) competition with ARL and TDAC, designing challenge scenarios and infrastructure that simulate real-world offensive and defensive exercises.
- Core research deliverable: ModelScope, a benchmarking platform for autonomous AI agents (detailed under Research Projects).

Cybersecurity Rapid Innovation Group (CyberRIG), UTEP Spring 2026 – Present
Research Assistant / Member

- Research applied defenses for autonomous AI systems, translating findings into prototype tools.
- Design and run workshops and hackathons that simulate real-world cybersecurity incidents for fellow students and researchers.

RESEARCH PROJECTS

ModelScope — Evaluation Platform for Autonomous AI Agents *(Army HBCU-MI SPARK, 2026)*

- Building a benchmarking platform that evaluates autonomous AI agents in complex, simulated environments.
- Designed a modular plugin architecture enabling configurable agents and objective ground-truth evaluation criteria.
- Captures real-time metrics on task success, operational speed, resource usage, and policy compliance for auditable validation of field-deployable AI systems.

SpecGuard — AI-Assisted Vulnerability Discovery System

- Built a hybrid vulnerability detection framework combining static analysis, LLM reasoning, and RAG over a ChromaDB knowledge base indexing the hzhu721/vulnerability-specifications dataset — 1,999 real-world CVEs — from Hugging Face.
- Implemented detection logic for memory corruption, use-after-free, and integer overflow vulnerability classes.
- Built a pairwise-correctness evaluation pipeline to score detection output against the CVE ground truth at scale.

Evacuation Optimization Simulator *(Advanced Algorithms coursework)*

- Developed a multi-agent evacuation simulation using graph-based modeling with congestion-aware Dijkstra and Floyd–Warshall routing algorithms.
- Modeled real-world constraints — edge capacity, congestion, variable movement speed — and analyzed results using evacuation rate, time-to-safety, and population throughput.

TECHNICAL SKILLS

Programming: Java, Python, SQL, PHP, JavaScript
AI & Data Systems: Retrieval-Augmented Generation (RAG), LangChain, ChromaDB, Ollama, autonomous-agent evaluation
Cybersecurity: Network Analysis, Authentication Systems, Vulnerability Assessment, Ethical Hacking, NIST Cybersecurity Framework
Tools & Systems: Linux, Wireshark, Git, secure web systems, object-oriented design

CERTIFICATIONS

- Google Cybersecurity Certificate (Coursera, 2025)
- Offensive Security Ethical Hacking Cert. — Cisco (In Progress)